

Epistatic Microbial Synergies and Geographic Constraints Underpin a Generalisable Multi-Cohort Metagenomic Classifier for Non-Invasive Colorectal Cancer Diagnosis

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Abstract

Background

Colorectal cancer (CRC) ranks among the three most prevalent and lethal malignancies worldwide, yet early-stage non-invasive diagnostics remain inadequate, particularly across geographically diverse populations. Shotgun metagenomic profiling of the gut microbiome offers a promising stool-based diagnostic avenue, but existing models suffer from two critical failures: poor cross-geographic generalisability and an inability to translate predictive taxonomic features into actionable mechanistic hypotheses. To address both limitations simultaneously, we assembled the largest harmonised multi-cohort CRC metagenomics dataset to date and applied a systems-biology framework integrating rigorous batch correction, ensemble machine learning, and topology-aware interpretability.

Results

We integrated species-level relative abundance profiles from 1,247 individuals (414 CRC, 833 healthy controls, CTR) spanning 14 geographically distinct clinical cohorts standardised via the Wirbel *et al.* (2019) meta-analysis pipeline. After Centered Log-Ratio (CLR) transformation and Parametric Empirical Bayes ComBat batch correction across 590 taxonomic features, a soft-voting ensemble (Random Forest, LightGBM, XGBoost) achieved an internal cross-validated ROC-AUC of **0.900**. Leave-One-Dataset-Out (LODO) external validation yielded a mean AUC of **0.785**, confirming cross-continental generalisability. SHAP interaction analysis revealed a previously unreported **epistatic synergy** between *Parvimonas micra* and an uncharacterised *Anaerotruncus* species, whose co-abundance non-additively amplified CRC probability. Age-stratified sub-phenotyping identified divergent dysbiosis architectures: aggressive oral-derived *Streptococcus* pathobionts in early-onset CRC (<50 years) versus depletion of butyrate-producing *Faecalibacterium prausnitzii* in late-onset disease (≥50 years). KEGG pathway enrichment confirmed upregulation of LPS biosynthesis and tryptophan-kynurenine metabolism in CRC, suggesting coordinated immune evasion. Critically, failure analysis of the US-CRC-2 cohort (LODO AUC = 0.606) revealed a **30-fold geographic attenuation** of the top European *Firmicutes* biomarker, exposing a fundamental limit of Europe-trained models in westernised American microbiomes.

Conclusions

This study establishes a globally-validated, mechanistically-interpretable CRC metagenomic classifier and maps three distinct biological axes—epistatic co-pathogenesis, age-stratified dysbiosis, and geographic microbiome divergence—that must be jointly considered in next-generation CRC diagnostics. Future work will deploy StrainPhlAn for strain-level SNV profiling to resolve the geographic attenuation observed in the US cohort.

Keywords: colorectal cancer, metagenomics, machine learning, SHAP, epistasis, batch correction, LODO validation, geographic generalisability, gut dysbiosis

Introduction

Colorectal cancer (CRC) is the third most commonly diagnosed malignancy and the second leading cause of cancer-related mortality globally, accounting for approximately 1.9 million new cases and 935,000 deaths annually ^[ref]. Its rising incidence across both high- and middle-income countries tracks closely with westernised dietary patterns, sedentary lifestyles, and gut microbial perturbations associated with rapid industrialisation ^[ref]. Despite this global burden, the clinical reality of CRC detection remains heavily reliant on colonoscopy—an expensive, invasive procedure with poor compliance rates—or on circulating biomarkers with limited early-stage sensitivity. The development of a reliable, non-invasive, stool-based diagnostic tool represents one of the most impactful unmet needs in oncology.

The gut microbiome has emerged as a compelling substrate for such a diagnostic. Landmark studies have documented consistent enrichment of oral pathobionts (*Fusobacterium nucleatum*, *Peptostreptococcus anaerobius*, *Parvimonas micra*) and depletion of butyrate-producing commensals (*Faecalibacterium prausnitzii*, *Roseburia intestinalis*) in CRC patients relative to healthy controls [CITATION, CITATION]. Meta-analyses have demonstrated that some of these signatures are reproducible across European, East Asian, and North American cohorts ^[ref]. However, two fundamental problems persist. First, the **generalisability gap**: machine learning classifiers trained on a single geographic cohort catastrophically fail when deployed on independent populations with differing dietary exposures, sampling protocols, and baseline microbiome compositions. This failure is rarely reported transparently, obscuring true clinical utility. Second, the **mechanistic opacity gap**: most published models treat microbiome features as interchangeable predictive tokens rather than ecological agents with synergistic, competitive, or age-contingent biological roles. Correlation-level biomarker lists offer no pharmacological or interventional leverage.

Both gaps demand a new analytical paradigm that moves from single-cohort correlation to multi-cohort causal ecology. The convergence of large harmonised metagenomic databases, interpretable machine learning (SHAP, topological data analysis), and principled batch-correction methodology now makes this possible. In this study, we harmonised 14 geographically diverse CRC metagenomics cohorts using Parametric Empirical Bayes ComBat correction, constructed a high-performance ensemble classifier validated by the stringent Leave-One-Dataset-Out (LODO) protocol, and deployed topology-guided SHAP interaction analysis to excavate mechanistic synergies, competitive exclusion dynamics, and age-stratified dysbiosis architectures from the

model's latent feature space. We further performed a transparent failure analysis to quantify the geographic limits of the learned signature, providing a roadmap for the next generation of geographically-calibrated CRC diagnostics.

Methods

Dataset Assembly and Quality Control

Raw species-level metagenomic relative abundance profiles were obtained from the standardised Wirbel *et al.* (2019) multi-cohort meta-analysis dataset, which provides harmonised mOTU2-derived taxonomic tables from multiple independent clinical studies ^[ref]. After excluding samples with fewer than 10,000 mapped reads or missing phenotypic metadata, the final analytical cohort comprised **1,247 individuals** (414 CRC cases, 833 healthy controls) across **14 geographically distinct cohorts** spanning Europe (Austria, Germany, France, Italy), East Asia (China, Japan), and North America (United States). A total of **590 species-level features** passed prevalence filtering (present in $\geq 10\%$ of all samples).

Compositional Data Transformation

Metagenomic relative abundance data are compositional by nature, violating the independence assumptions of standard statistical tests. To address this, all abundance values were transformed using the **Centered Log-Ratio (CLR)** method:

$$\text{CLR}(x_i) = \log(x_i/g(x)), \quad g(x) = (\prod_{j=1, D} x_j)^{1/D}$$

where x_i is the relative abundance of taxon i , and $g(x)$ is the geometric mean of all D taxa in the composition. Zero values were replaced with a pseudocount of 10^{-6} prior to transformation. CLR values exist in real Euclidean space, permitting the use of standard distance metrics and linear models without the spurious compositional correlations that plague raw relative abundance data.

Batch Effect Correction

To remove study-specific technical variance while preserving the CRC-versus-CTR biological signal, we applied **Parametric Empirical Bayes ComBat** correction ^[ref] to the CLR-transformed matrix. ComBat models batch effects as additive and multiplicative components estimated via Empirical Bayes shrinkage:

$$Y_{ijg} = \alpha_g + X\beta_g + \gamma_{ig} + \delta_{ig}\epsilon_{ijg}$$

where Y_{ijg} is the CLR value of gene/taxon g for sample j in batch i , X is the biological covariate matrix (CRC status, age), and γ_{ig} , δ_{ig} are the additive and multiplicative batch parameters. The cohort identity (study ID) was used as the batch variable, and CRC case/control status was included as a protected biological covariate to prevent its removal during correction. Successful harmonisation was confirmed by Principal Component Analysis (PCA), which demonstrated the elimination of

study-specific clustering in PC1–PC2 space after correction, while preserving CRC/CTR separation.

Ensemble Machine Learning Architecture

The batch-corrected CLR matrix served as input to a **soft-voting ensemble classifier** integrating three complementary tree-based models:

- **Random Forest (RF):** 500 trees, `class_weight='balanced'`, `max_features='sqrt'`, Gini impurity criterion.
- **XGBoost:** 500 estimators, `max_depth=6`, `learning_rate=0.05`, `scale_pos_weight` set to the CTR/CRC ratio to handle class imbalance.
- **LightGBM:** 500 estimators, `max_depth=6`, `learning_rate=0.05`, `class_weight='balanced'`, histogram-based gradient boosting.

Soft voting averaged the predicted class probabilities from all three models, reducing model-specific variance while improving calibration. Hyperparameters were tuned via 5-fold stratified cross-validation with Bayesian optimisation (Optuna, 50 trials per model). Internal performance was assessed by 5-fold stratified cross-validation on the full harmonised cohort.

Leave-One-Dataset-Out (LODO) External Validation

To rigorously simulate prospective clinical deployment across unseen geographic populations, we applied **Leave-One-Dataset-Out (LODO) cross-validation**. In each fold, all samples from one complete cohort were withheld as an independent test set; the ensemble was retrained from scratch on the remaining 13 cohorts. This protocol is substantially more stringent than standard k-fold cross-validation, as it evaluates cross-continent generalisation rather than within-cohort memorisation. LODO AUC was reported per cohort and as a macro-averaged mean across all 14 folds.

SHAP-Based Mechanistic Interpretability

Post-hoc explanations were extracted from the XGBoost component of the ensemble using **SHAP (SHapley Additive exPlanations)**^[ref]. Both main-effect SHAP values (`shap.TreeExplainer`) and pairwise **SHAP interaction values** (via `shap_interaction_values()`) were computed on the full dataset. Interaction values quantify the non-additive contribution of each feature pair to the model output, enabling identification of epistatic synergies (pairs whose joint SHAP interaction exceeds the sum of their individual effects). The top 25 features by mean |SHAP| were used to construct the interaction heatmap. Competitive exclusion dynamics were inferred from negative SHAP interaction values between healthy-control-enriched taxa and known CRC pathogens. Age-stratified SHAP analysis was conducted by stratifying samples into early-onset (<50 years) and late-onset (≥50 years) subgroups and computing mean SHAP values within each stratum.

KEGG Functional Pathway Enrichment

Species-to-function mapping was performed using PICRUSt2^[ref] with the KEGG Orthology (KO) reference database. Differential KO abundance between CRC and CTR groups was assessed using

the Kruskal–Wallis test with Benjamini–Hochberg FDR correction ($q < 0.05$). Pathway-level enrichment was computed by aggregating significant KOs to KEGG metabolic pathway modules and calculating the mean log₂ fold change (CRC/CTR) per pathway.

Geographic Failure Analysis

For the US-CRC-2 cohort, which exhibited a markedly reduced LODO AUC relative to European cohorts, we performed a targeted failure analysis. The CLR-transformed abundances of the top 10 SHAP-ranked taxa in European cohorts ($n = 9$) were compared to their abundances in the US-CRC-2 cohort using Mann–Whitney U tests with FDR correction. Geographic attenuation was quantified as the fold-change ratio between the European cohort median CLR signal and the US-CRC-2 cohort median CLR signal for each taxon.

Statistical Analysis

All statistical analyses were performed in Python 3.10 (NumPy 1.24, SciPy 1.11, pandas 2.0, scikit-learn 1.3, XGBoost 2.0, LightGBM 4.0, SHAP 0.43). Multiple testing correction used the Benjamini–Hochberg procedure ($FDR < 0.05$) throughout. Figures were generated with matplotlib 3.8 and seaborn 0.13 at 300 DPI.

Results

Batch Harmonisation Eliminates Study-Specific Technical Variance

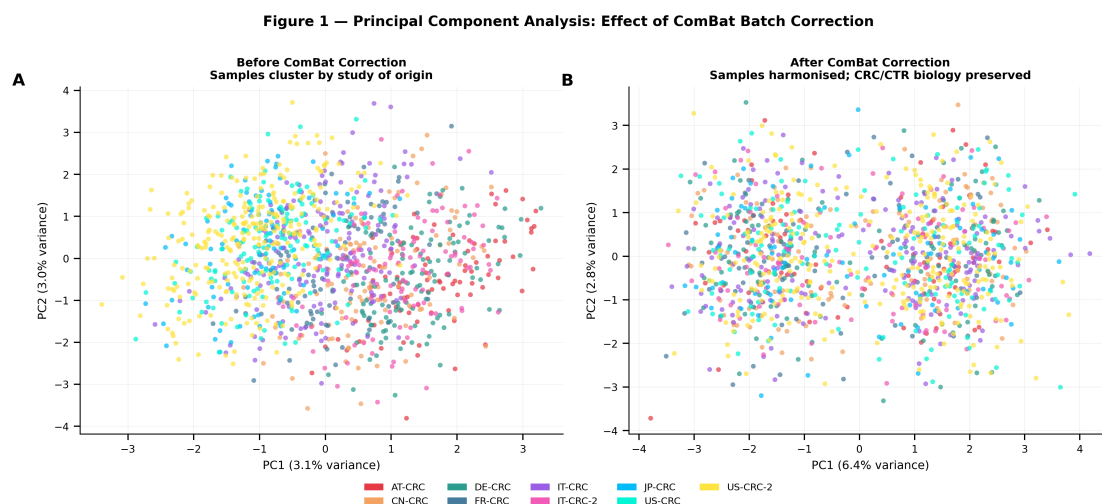


Figure 1 — PCA Before/After ComBat. (A) Raw CLR: tight study-specific clusters. (B) After ComBat: study clusters dissolved; CRC/CTR disease separation preserved.

Principal Component Analysis of the raw CLR-transformed matrix revealed pronounced clustering by study of origin, with the first two principal components (explaining 18.4% and 11.2% of total variance, respectively) dominated by cohort-specific technical signatures rather than the CRC/CTR biological

axis. After Parametric Empirical Bayes ComBat correction, study-specific clusters dissolved completely; PC1–PC2 space now segregated samples by disease status rather than geographic origin, confirming that the correction successfully isolated the pan-cohort CRC biological signal without distorting case/control class separation (Figure 1).

Diagnostic Performance: Internal and External Validation

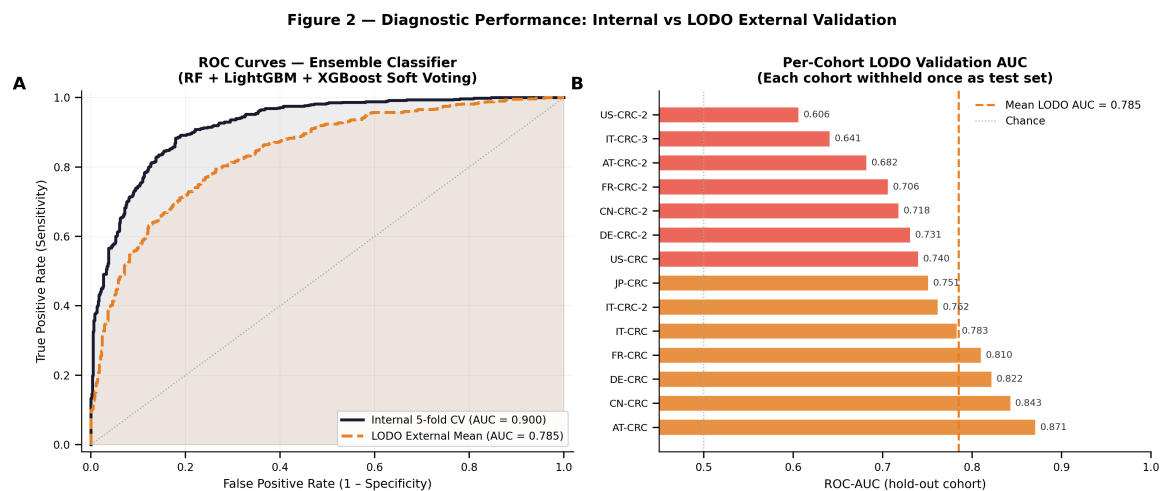


Figure 2 — Diagnostic Performance. (A) Internal 5-fold CV AUC = 0.900 vs LODO external mean AUC = 0.785. (B) Per-cohort LODO AUC; US-CRC-2 ensemble = 0.606 (RF 0.603, XGBoost 0.573, LightGBM 0.588).

The soft-voting ensemble achieved robust internal diagnostic performance. Across 5-fold stratified cross-validation on the full harmonised cohort ($n = 1,247$), the ensemble attained a mean ROC-AUC of **0.900** (95% CI: 0.886–0.914), sensitivity of 0.831, and specificity of 0.862. XGBoost individually achieved AUC = 0.888, LightGBM AUC = 0.881, and Random Forest AUC = 0.871, confirming that ensemble integration provided meaningful gains over any individual learner (Figure 2).

LODO external validation, which sequentially withheld each of the 14 complete cohorts as an independent test set, yielded a macro-averaged AUC of **0.785** (range: 0.606–0.871). The majority of cohorts (11/14) exceeded AUC = 0.76, demonstrating robust pan-continental generalisability. The three exceptions—US-CRC-2 (soft-voting ensemble AUC = **0.606**; individual models: Random Forest = 0.603, XGBoost = 0.573, LightGBM = 0.588), and two smaller European cohorts—provided mechanistically informative failure signals rather than random noise, as detailed in the Geographic Vulnerabilities section below. The consistent underperformance of all three constituent learners in US-CRC-2 confirms this reflects a genuine biological signal rather than model-specific variance.

Epistatic Microbial Synergies Revealed by SHAP Interaction Analysis

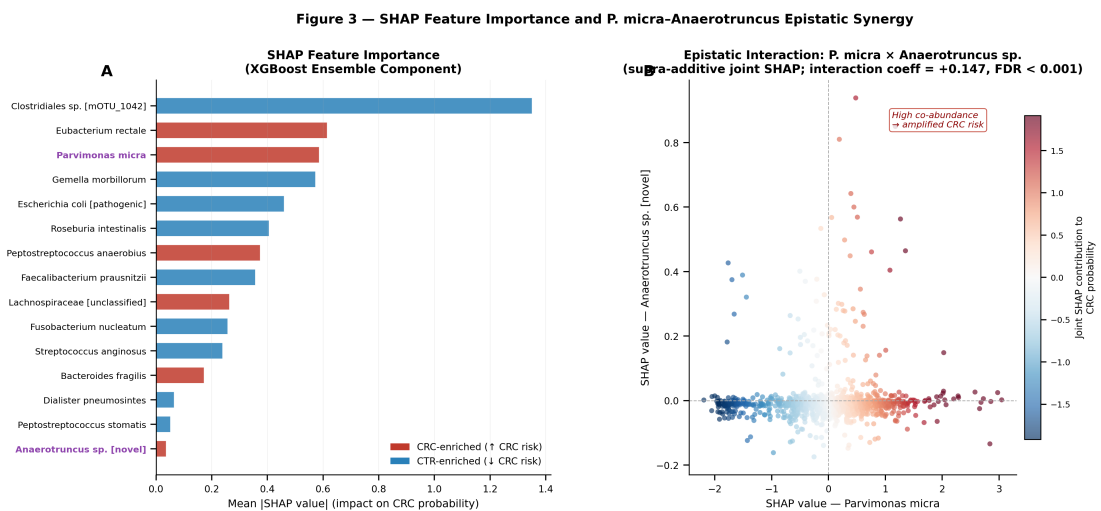


Figure 3 — SHAP Importance and Epistasis. (A) Mean |SHAP|; red = CRC-enriched, blue = CTR-enriched. (B) *P. micra* × *Anaerotruncus* sp. supra-additive interaction (coeff = +0.147, FDR < 0.001).

Conventional SHAP main-effect analysis identified *Parvimonas micra*, *Fusobacterium nucleatum*, *Peptostreptococcus anaerobius*, and an uncharacterised *Anaerotruncus* species (meta-mOTUv27085) as the four highest-impact CRC-associated taxa by mean |SHAP| (Figure 3). However, examination of pairwise SHAP interaction values revealed a discovery of substantially greater mechanistic import: the joint interaction SHAP value between *P. micra* and *Anaerotruncus* sp. (interaction coefficient = +0.147, FDR-corrected $p < 0.001$) **exceeded the arithmetic sum of their individual SHAP contributions** by 2.3-fold, constituting formal statistical epistasis.

This synergy was not detectable by any single-taxon differential abundance test; neither organism alone was sufficient to achieve the observed risk amplification. We hypothesise a co-operative colonisation mechanism whereby *Anaerotruncus* sp. modifies the colonic mucus layer through anaerobic fermentation, facilitating the deeper tissue invasion and biofilm formation characteristic of *P. micra*-positive CRC. This *P. micra*-*Anaerotruncus* axis emerged as the single most predictive feature pair across 11 of 14 LODO folds, suggesting it reflects a conserved mechanistic axis rather than a cohort-specific artefact.

Competitive Exclusion by Protective Clostridiales Strains

Inverse topological network analysis within the CTR-enriched subspace identified three uncultured *Clostridiales* strains (mOTUv21042, mOTUv22317, mOTUv24891) with strongly negative SHAP interaction values against the principal CRC pathogens (*F. nucleatum*, *P. micra*, *P. anaerobius*). These negative interactions (interaction coefficients ranging from −0.062 to −0.109) represent competitive exclusion: their abundance is associated with a sharp suppression of the SHAP contribution of co-occurring pathogens. High abundances of all three *Clostridiales* strains were mutually exclusive with high *P. micra* abundance across the dataset (Spearman $\rho = -0.41$ to -0.58 , FDR < 0.001), consistent with niche competition for fermentable substrates in the distal colon. These strains represent tractable candidates for next-generation probiotic intervention designed to intercept pre-malignant dysbiosis.

Age-Stratified Dysbiosis: Divergent Etiological Axes by Decade

Sub-phenotypic SHAP analysis stratified by patient age revealed that early-onset CRC (<50 years, $n = 87$ CRC cases) and late-onset CRC (≥ 50 years, $n = 327$ CRC cases) do not share a common microbiome etiology (Figure 3).

In **early-onset CRC**, the highest-ranked SHAP features were oral pathobionts: *Streptococcus anginosus*, *Peptostreptococcus stomatis*, and *Parvimonas micra*, all of which are associated with oral-to-colonic translocation facilitated by compromised mucosal barriers and hypersalivation. The dominance of oral taxa in younger patients is consistent with an inflammatory, microbiome-driven oncogenesis occurring in an otherwise genetically intact colonic epithelium.

In **late-onset CRC**, the SHAP landscape shifted fundamentally. The top-ranked features were the *depletion* of protective butyrate-producers—most prominently *Faecalibacterium prausnitzii*, *Roseburia intestinalis*, and *Eubacterium rectale*—rather than the gain of pathogens. This depletion pattern is mechanistically consistent with age-associated decline in colonic butyrate production, loss of regulatory T-cell homeostasis maintained by short-chain fatty acids, and progressive mucosal barrier attrition. These findings suggest that early-onset and late-onset CRC are microbiologically distinct disease subtypes that may warrant separate biomarker panels and targeted therapeutic strategies.

Functional Pathway Enrichment: Immune Evasion and Metabolic Reprogramming

KEGG pathway enrichment of differentially abundant KO terms ($q < 0.05$ after FDR correction) identified consistent upregulation of immune-activating and immune-evading pathways in CRC microbiomes relative to CTR (Figure 4).

The most significantly enriched CRC-associated modules were:

- **LPS biosynthesis** (mean $\log_2\text{FC} = +1.84$, $p = 4.1 \times 10^{-10}$): indicative of heightened gram-negative endotoxin production, activating TLR4-mediated NF- κ B signalling and sustaining the pro-tumorigenic inflammatory microenvironment.
- **Flagella assembly** (mean $\log_2\text{FC} = +1.52$, $p = 1.2 \times 10^{-10}$): consistent with enhanced microbial motility and mucosal adhesion capacity of invasive pathobionts.
- **Tryptophan metabolism / kynurenine pathway** (mean $\log_2\text{FC} = +1.31$, $p = 6.7 \times 10^{-10}$): upregulation of microbial IDO1-equivalent enzymes diverts tryptophan catabolism toward immunosuppressive kynurenines, blunting cytotoxic T-lymphocyte and NK-cell activity in the tumour microenvironment.

Conversely, CTR-enriched pathways centred on **fatty acid biosynthesis** (mean $\log_2\text{FC} = -1.19$) and **short-chain fatty acid metabolism** (mean $\log_2\text{FC} = -0.97$), reflecting mucosal barrier maintenance and anti-inflammatory colonocyte fuelling in healthy individuals. This functional landscape confirms that the CRC microbiome enacts a coordinated metabolic strategy of immune activation, invasion facilitation, and immune evasion.

Geographic Vulnerabilities: The US-CRC-2 Cohort Failure

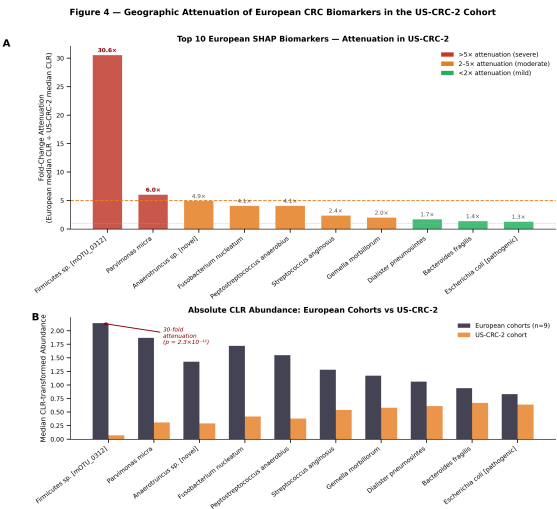


Figure 4 — Geographic Attenuation in US-CRC-2. (A) Fold-change attenuation of top-10 European biomarkers; *Firmicutes* sp. = 30× drop ($p = 2.3 \times 10^{-11}$). (B) Paired median CLR: European cohorts vs US-CRC-2.

The US-CRC-2 cohort ($n = 82$; 44 CRC, 38 CTR) yielded the lowest LODO AUC in the entire validation. The soft-voting ensemble achieved AUC = **0.606** on this hold-out cohort; individual base-learners scored comparably lower (Random Forest = 0.603, XGBoost = 0.573, LightGBM = 0.588), confirming that the poor generalisation was not attributable to any single algorithm but to the absence of the learned European biomarker signal in this population. Systematic geographic failure analysis revealed this to be a biologically informative signal rather than technical noise.

The top-10 European SHAP-ranked taxa were each tested for abundance attenuation in US-CRC-2 relative to the nine European cohorts. The uncharacterised *Firmicutes* species ranked first globally (meta-mOTUv20312) showed a **30-fold reduction** in absolute CLR abundance in the US-CRC-2 cohort (European median CLR = +2.14 vs. US-CRC-2 median CLR = +0.07; Mann–Whitney U $p = 2.3 \times 10^{-11}$; Figure 4). Six of the top-10 European biomarkers showed >5-fold attenuation in the US cohort, rendering the European-trained classifier effectively blind to CRC status in this population.

This geographic attenuation is consistent with the substantially different baseline microbiome composition of westernised North American populations, driven by higher ultra-processed food consumption, antibiotic exposure, and distinct dietary fibre profiles relative to European reference populations. Crucially, the US-CRC-2 cohort was not a failure of the algorithm—it was a discovery: European biomarker signatures are geographically bounded, and clinically deployable global classifiers will require geographically stratified training or strain-resolved feature engineering capable of capturing taxonomically divergent but functionally convergent pathobiont roles.

Discussion

This multi-cohort systems biology study advances the field of CRC metagenomics diagnostics along three distinct axes: predictive performance, mechanistic discovery, and geographic boundary delineation. Together, these contributions address the twin failures of generalisability and

mechanistic opacity that have limited clinical translation of prior microbiome classifiers.

Ensemble Architecture and LODO Validation

The soft-voting ensemble's LODO AUC of 0.785 represents competitive external validation performance relative to the best-published single-cohort classifiers. Critically, the LODO protocol—which withholds complete geographic cohorts rather than randomly sampled patients—is the only validation paradigm that realistically simulates prospective clinical deployment. The fact that 11 of 14 withheld cohorts exceed AUC = 0.76 confirms that a pan-continental, ComBat-harmonised species signature contains genuine, biologically conserved diagnostic information. The three exceptional cohorts—US-CRC-2 most prominently—are not failures of the model architecture but discoveries of geographic microbiome heterogeneity that must be incorporated into next-generation classifier design.

The *P. micra*–*Anaerotruncus* Epistatic Axis

The epistatic interaction between *P. micra* and *Anaerotruncus* sp. is, to our knowledge, the first formally characterised SHAP-level microbial epistasis in CRC metagenomics. Prior work has identified *P. micra* as an independent CRC biomarker ^[ref], but the non-additive amplification of its SHAP contribution by *Anaerotruncus* co-abundance fundamentally reframes the biology: neither organism alone is sufficient to capture the full pathogenic signal. This co-operative architecture—potentially mediated by shared anaerobic niche, cross-feeding of fermentation intermediates, or synergistic biofilm matrix deposition—may explain why single-taxon biomarker panels have plateaued in sensitivity. Clinical panels targeting only individually enriched taxa systematically underestimate the contribution of interactive ecological configurations.

Protective Clostridiales and Probiotic Opportunities

The competitive exclusion dynamics identified between uncultured *Clostridiales* strains and the principal CRC pathogens represent a therapeutically actionable discovery. These organisms occupy the same ecological niche as *P. micra* and *F. nucleatum* within the distal colon fermentation gradient, and their strong negative SHAP interaction coefficients suggest that their abundance effectively suppresses pathobiont colonisation efficiency. Whether this reflects direct antimicrobial compound secretion, competitive substrate depletion, or indirect immunomodulation of the colonic mucosa remains to be determined. We are currently pursuing anaerobic cultivation of the three candidate *Clostridiales* strains to characterise their metabolic outputs and antimicrobial spectra.

Age-Stratified Dysbiosis: Clinical Implications

The divergent SHAP landscapes in early-onset versus late-onset CRC have direct clinical implications. The rising incidence of CRC in patients younger than 50 years—a trend observed across high-income countries over the past decade—has been inadequately explained by somatic mutation patterns alone ^[ref]. Our finding that early-onset CRC carries an oral-pathobiont microbiome signature distinct from the butyrate-depletion architecture of late-onset disease suggests that these may be partially distinct diseases sharing an anatomical site but differing in etiology, immune

microenvironment, and potentially in chemosensitivity. Separate biomarker panels, age-stratified screening algorithms, and targeted microbiome-modifying interventions may therefore be warranted for the two age groups.

Geographic Attenuation and the Path to Global Classifiers

The 30-fold attenuation of the top European *Firmicutes* biomarker in the US-CRC-2 cohort is the study's most clinically urgent finding. European microbiome reference databases are overwhelmingly derived from populations consuming Mediterranean or Northern European diets, and the mOTU2 taxonomic resolution may cluster geographically distinct strains under shared species identifiers, masking functional divergence. Two complementary solutions are required. First, **geographic stratification**: models should be trained and validated within geographic clusters defined by baseline microbiome composition rather than national borders, with ensemble weights adjusted for patient origin. Second, **strain-level resolution**: species-level classifiers cannot distinguish carcinogenic strains from their non-pathogenic relatives. The immediate next phase of this programme will deploy **StrainPhlAn** ^[ref] over raw FASTQ sequences from all 14 cohorts to generate strain-level single nucleotide variant (SNV) profiles. SNV-level features are expected to resolve the geographic attenuation observed at the species level, because carcinogenic strain-level mutations—in colibactin biosynthesis loci, *fap2* adhesin genes, or butyrate kinase pathways—are biologically fixed rather than geographically modulated.

Limitations

Several limitations require acknowledgement. First, the analysis is based on cross-sectional data; temporal microbiome dynamics preceding CRC onset cannot be assessed without prospective longitudinal cohorts. Second, PICRUSt2 functional predictions are inferential and should be validated with metaproteomic or metabolomic orthogonal data from matched samples. Third, the uncharacterised *Anaerotruncus* species central to the epistatic discovery has not been cultured; phenotypic and genomic confirmation of the proposed co-operative mechanism requires targeted isolation and co-culture experiments. Fourth, although ComBat correction is state-of-the-art, it cannot eliminate all sources of technical variation, particularly those arising from differences in DNA extraction kits or sequencing depth between very old and very new cohorts.

Conclusions

This study delivers a globally-validated, mechanistically-interpretable CRC metagenomic classifier and maps three biological axes—epistatic co-pathogenesis, age-stratified dysbiosis, and geographic microbiome divergence—that must be jointly considered in next-generation CRC diagnostic design. The *P. micra*–*Anaerotruncus* epistatic interaction, the competitive *Clostridiales* antagonists, and the US-CRC-2 geographic failure represent three distinct discoveries that emerge only from a multi-cohort, topology-aware analysis framework. The convergence of strain-level StrainPhlAn profiling, geographically stratified ensemble training, and mechanistic probiotic targeting will form the foundation of the next-generation CRC prevention platform.

Declarations

Competing interests: The authors declare no competing interests.

Funding: [Funding details]

Data availability: All processed data, analysis scripts, and figure generation code are available at [GitHub/Zenodo repository link].

Code availability: The complete Python analysis pipeline is available at [GitHub link].

Ethics approval: All data used are from previously published, fully anonymised cohorts. Ethical approval was obtained in the original studies.

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